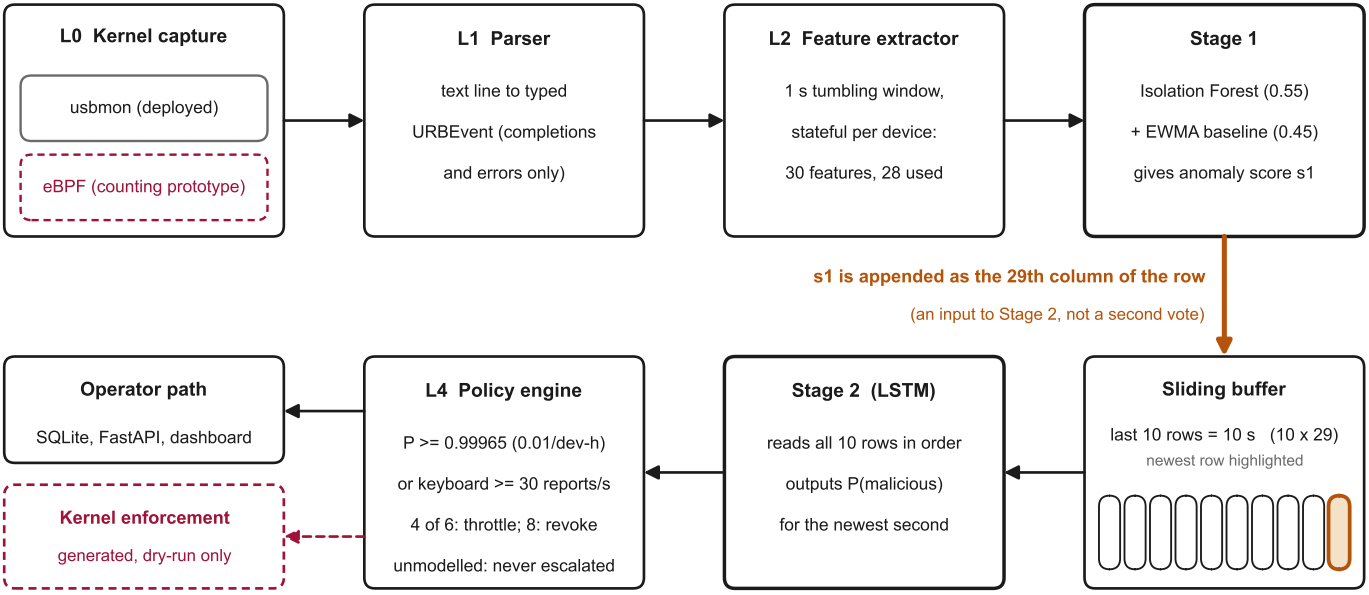
L0 Kernel capture
usbmon (deployed)
eBPF (counting prototype)
L1 Parser
text line to typed
URBEvent (completions
and errors only)
L2 Feature extractor
1 s tumbling window,
stateful per device:
30 features, 28 used
Stage 1
Isolation Forest (0.55)
+ EWMA baseline (0.45)
gives anomaly score s1
s1 is appended as the 29th column of the row
(an input to Stage 2, not a second vote)
Sliding buffer
last 10 rows = 10 s (10 x 29)
newest row highlighted
Stage 2 (LSTM)
reads all 10 rows in order
outputs P(malicious)
for the newest second
L4 Policy engine
P >= 0.99965 (0.01/dev-h)
or keyboard >= 30 reports/s
4 of 6: throttle; 8: revoke
unmodelled: never escalated
Operator path
SQLite, FastAPI, dashboard
Kernel enforcement
generated, dry-run only